UMSL Mechanical Engineering Course Categories

School of Engineering, 2025-2026 Catalog

Engineering Core Courses (27 credits)

Course	Title	Cr	Description
ENGR 1414	Elementary Engineering Design	2	Introduction to engineering design process, project planning, teamwork, and systems integration through design-build-test projects
ENGR 2310	Statics	3	Equilibrium of particles and rigid bodies; distributed forces, centroids, friction, and structural analysis
ENGR 2320	Dynamics	3	Kinematics and kinetics of particles and rigid bodies; work-energy and impulse-momentum methods
ENGR 2330	Introduction to Thermodynamics	3	Properties of substances, first and second laws of thermodynamics, entropy, ideal gas applications
ENGR 2332	Mechanics of Materials	3	Stress and strain in structural members: axial, torsion, bending, shear, combined loading, deflection
ENGR 3300	Applied Thermodynamics	3	Power and refrigeration cycles, gas mixtures, psychrometrics, combustion, exergy analysis
ENGR 2022	Engineering Economics and PM	3	Time value of money, economic analysis methods, depreciation, project management fundamentals
EENG 2310	Circuit Analysis I	3	DC and AC circuit analysis; Kirchhoff's laws, network theorems, transients, phasors, frequency response
CMP SCI 1250	Introduction to Computing	3	Python programming for engineers; data analysis, visualization, computational problem solving
ENGR 4400	FE Exam Review	1	Comprehensive review of all engineering topics for the Fundamentals of Engineering licensing exam

ME Core Courses (24 credits)

Course	Title	Cr	Description
MENG 1204	ME 3D Design (CAD)	2	Introduction to CAD using SolidWorks; 2D sketching, 3D solid modeling, assemblies, engineering drawings
MENG 3330	Instrumentation and Measurement	3	Measurement statistics, sensors (temperature, pressure, force, flow), signal conditioning, data acquisition
MENG 3340	Properties of Material and Testing	3	Crystal structure, phase diagrams, mechanical properties, failure modes, materials selection, testing methods
MENG 3350/L	System Dynamics and Control + Lab	4	System modeling, Laplace transforms, transfer functions, PID control, frequency response; lab experiments
MENG 3360	Machine Design and Manufacturing	3	Design of machine elements (shafts, bearings, gears, springs); fatigue analysis, manufacturing processes, DFM
MENG 3370/L	Fluid Mechanics + Lab	4	Fluid properties, statics, Bernoulli equation, conservation laws, dimensional analysis, pipe flow, external flow
MENG 3380/L	Heat Transfer + Lab	4	Steady/transient conduction, fins, forced/natural convection, radiation, heat exchangers
MENG 3390	Product Dev. and Prototyping Lab	1	Product development process, additive manufacturing, reverse engineering, rapid prototyping

ME Advanced / Elective Courses (10 credits)

Course	Title	Cr	Description
MENG 4440	Intro to Composite Materials	3	Fiber/matrix materials, micromechanics, laminate theory, failure criteria, manufacturing, testing
MENG 4450	Computational Fluid Dynamics	3	Governing equations, discretization, grid generation, turbulence models, commercial CFD software
MENG 4460	Renewable Energy Systems Lab	1	Hands-on lab with solar, wind, and fuel cell systems; data acquisition and efficiency analysis
MENG 4490	Advanced Manufacturing	3	CNC machining, advanced additive manufacturing, automation, Industry 4.0, smart manufacturing

Capstone / Design Courses (4 credits)

Course	Title	Cr	Description
MENG 4980	Senior Design I	2	First capstone: project planning, concept generation/selection, market analysis, preliminary design
MENG 4990	Senior Design II	2	Second capstone: detailed analysis, fabrication, testing, validation, documentation, final presentation